

Communication, understanding and engagement of patients with pancreatic cancer at time of diagnosis

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ABSTRACT

Background: Objectives: To investigate communication clarity and understanding at the time of pancreatic adenocarcinoma (PDAC) diagnosis and whether they can influence patient engagement and compliance.

Methods: Consecutive PDAC patients were enrolled at the time of diagnosis after obtaining informed consent in a single-center study. The patients completed a validated scale (PHE-s®), and the understanding rate was assessed using standardized tools. Patient compliance was evaluated, and the correlation between the PHE-s®, understanding, and compliance was calculated.

Results: Thirty patients were enrolled (15 female) with a mean age 64.4, 13 were metastatic. The mean visit time was 31 min, being longer if visiting doctor was an oncologist ($p = 0.002$). The engagement level was high in 70% of the patients, and all but one were compliant. The analysis of doctor-patient interactions showed a median of 121 conversational turns for doctors, 75 for patients, and 20 for caregivers ($p < 0.0001$), and the median percentage of speaking time was 77% for doctors, 13% for patients, and 2% for caregivers ($p < 0.0001$). Female caregivers spent more time speaking than did male caregivers (median 11.6% vs. 1.3%; $p = 0.06$). There were 290 instances of problematic understanding, most of which occurred during the taking of patients' personal medical history for doctors, while for patients and caregivers, these occurred mainly during the discussion of diagnosis/treatment ($p < 0.0001$). In a multivariable analysis, only origin from central or southern Italy was associated with high engagement ($p = 0.0087$).

Conclusion: In this first attempt to measure clarity of communication and engagement in patients with PDAC, typical features of conversation and problematic understanding emerged, which deserves further investigation.

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1. Introduction

Trust between doctors and patients is crucial. Clear and correct communication constitutes the foundation for building a therapeutic alliance between the treating physician and the patient,

which is essential to ensure adherence to the proposed treatments [1]. This is particularly important for oncological care. Usually, patients trust physicians who are perceived as competent, but complex medical skills involved in dealing with cancer diagnostics and treatment cannot be easily evaluated by a patient [2]. Hence, the relationship of trust may depend on hard-to-quantify factors, including successful communication, especially during initial visits. However, few studies have been conducted on doctor-patient communication in cancer patients [3,4], and few have specifically focused on pancreatic ductal adenocarcinoma (PDAC) [5–8].

PDAC is a particularly aggressive tumour type, with a dismal prognosis due both to diagnosis typically reached at advanced stages and to poor response to available treatments. These factors lead to death in most patients, with a 5-year survival rate of 11% [9]. In the past few years, the level of patient engagement has been increasingly investigated as an important factor contributing to a solid therapeutic alliance: a strong and collaborative partnership between a healthcare provider (e.g., physician, nurse) and patient. This alliance is characterized by trust, effective communication, shared treatment goals, and mutual respect, with the aim of optimizing patient care and health outcomes [10,11]. In this regard, objective measurement scales of patients' engagement have been developed and demonstrated to be useful tools to quantify patient involvement in the care process [12,13].

We hypothesized that the clarity of communication between the treating physician and PDAC patients and their level of understanding, particularly at the time of diagnosis, can influence the construction of a solid therapeutic alliance and ultimately the patient's engagement and compliance with the proposed treatments.

Therefore, we designed this study to a) collect quantitative data about the level of PDAC patients' engagement; b) investigate the features of communication between doctors and patients with PDAC and the rate of patients' understanding of the information received from doctors; and c) investigate the associations between patients and disease-related variables, communication and understanding, and the level of patients' engagement and compliance.

2. Materials and methods

2.1. Study design, participants and data collection

The single center cross-sectional study named “Communication and Patient Engagement at Diagnosis of Pancreatic Cancer” was performed at IRCCS San Raffaele Research Hospital, Milan, Italy, from February 2021 to April 2022. The study was conducted in accordance with the ethical standards of the Declaration of Helsinki and approved by the Ethical Committee of San Raffaele Hospital (Communi.CARE; 52/INT/2019, amended November 11, 2020). All the participants provided informed consent. The inclusion criteria were as follows: a) outpatient visit in San Raffaele Hospital (either in the Gastroenterology/Pancreatic Disorders, Pancreatic Surgery, or Pancreatic Oncology Clinic) to communicate diagnosis and discuss the initial treatment plan for PDAC after histological confirmation in patients who were, therefore, naïve to cancer treatments; b) new diagnosis of the disease (within 4 weeks from histology); c) age > 18 years; d) Italian native language speakers; and e) willingness and ability to participate. The exclusion criteria were as follows: a) diagnosis of another tumor type or lack of histological confirmation, b) recurrence or progression of PDAC after primary treatment (not a new diagnosis), and c) poor performance status (Eastern Cooperative Oncology Group (ECOG) ≥ 3 [14]). The study protocol was recorded at ClinicalTrials.gov (NCT04257955), which was previously published [15] and included the following phases:

Patient selection and accrual: At the start of the visit and before

the communication of the histological diagnosis and treatment plan, the physician checked the inclusion and exclusion criteria and asked for oral consent. Recording of the visit: The protocol was initially approved before the COVID pandemic and contemplated the presence of a communication specialist, a Ph.D. student in Philosophy of Language and Communication (M.C.), who would have recorded the audio of the visits and took notes about the non-verbal clues that could have potentially been missed only using the audio recordings. Due to the pandemic, as the presence of non-essential, non-medical personnel was not allowed, the protocol was amended, and audio recording of the visit was performed by the treating physician upon the patient's consent. Given the clinical relevance of the communication between the diagnosis and treatment plan and the stressful event, the presence of one caregiver (who could be a spouse, partner, family member, or friend) was permitted upon request by the patient. During the visit, as per the normal procedure, the patients and caregivers could ask for clarification. In keeping with ethical approval, informed consent forms that included the term “adenocarcinoma” (cancer) were presented and signed by the patients at the end of the visit rather than at the beginning to avoid anticipating the diagnosis suddenly.

Evaluation of the patient engagement level: At the end of the visit, the patient completed the PHE-s[®] scale questionnaire. This allowed us to measure patient engagement, which encompasses an interactive and collaborative association between health care practitioners (e.g., physicians and nurses) and patients. This multifaceted concept underscores the significance of engaging patients as active participants in their healthcare decision-making processes, prioritizing transparent and open channels of communication, and nurturing a sense of empowerment and partnership between healthcare professionals and individuals seeking care. Medical engagement posits that patients should not merely assume passive roles in the healthcare encounter but should instead engage actively in the management of their health and the formulation of their treatment strategies, with healthcare providers assuming a supportive and facilitating role. The overarching objective of medical engagement is to increase patient satisfaction, enhance treatment adherence, and ultimately contribute to improved healthcare outcomes [16].

Follow-up interview from the communication specialist to evaluate understanding: At the outset, the study protocol involved informed consent to include a follow-up interview from the communication specialist (M.C.) to evaluate understanding immediately after the clinical consultation with the doctor had been completed, and the doctor had departed. As per the protocol amendments due to the pandemic, upon specific consent, the patient was instead reached over the phone within 72 h from the visit by the M.C. for the same interview. M.C. delivered a semi-structured interview to the patient to evaluate the clarity of the received information and ask about their perception of the possible misunderstandings that occurred during the visit. We conducted a qualitative analysis of these interviews, focusing on gaining insight into patients' perspectives on the challenges in their communication with physicians.

Recorded variables: The following variables were recorded during the visit by the physicians and during the interview by M.C. in a dedicated anonymized Case Report Form (CRF): age, sex, region of origin, educational level of the patient, date of the histologic diagnosis of PDAC, its stage, and the proposed treatment plan. All but six patients were at a visit with one or more caregivers (friends, spouse, partner, etc.).

2.2. Patient compliance and clinical outcomes

Patients were considered compliant if they adhered to the

treatment plan which was initially proposed, and non-compliant if they refused it, according to the assumption that compliance was defined as the patient's adherence to the prescribed treatment plan, medical advice, or recommended health behaviours as provided by healthcare professionals. Indeed, compliance generally involves the extent to which a patient follows the instructions and recommendations provided by their healthcare provider in terms of taking medications, attending appointments, following dietary or lifestyle recommendations, and engaging in other aspects of healthcare management. The overall survival of the patients at the end of the study was also recorded.

2.3. Patient health engagement: the PHE-s®

The Patient Health Engagement Scale (PHE-s®) is a validated assessment scale for simple and non-invasive use, which only requires the administration of a questionnaire [17]. The scale is based on the PHE model, a developmental psychological theory describing a patient's experience of becoming an active player in their healthcare pathway. This model describes four incremental positions of engagement: (1) blackout, (2) arousal, (3) adhesion, and (4) eudaimonic project. The first position ("blackout") is complete disengagement, which typically occurs when people feel vulnerable, psychologically frozen, and behaviourally paralyzed. It follows the psychological position of "arousal," in which people have acquired an initial awareness about their health situation but still do not have enough knowledge and skills to manage it. They appear hypervigilant over their body signals, hyperactive, and confused when seeking information about their health situation. It follows the position of "adhesion," in which patients have matured a good psychological adaptation to their health condition and appear more motivated to comply with medical prescriptions. Finally, when people mature to a complete awareness and acceptance of their illness conditions, they reach the "eudaimonic project" phase, with an increased ability to deal with the uncertainty of the moment and a strong motivation to psychologically achieve the sense of hope and a "new normality."

Specifically, patients were considered to have a "low PHE level" when they are positioned in Blackout or Arousal (1 or 2), whereas they were defined as "high PHE level" when their scoring was related to the Adhesion and Eudaimonic (3 or 4) positions.

2.4. Semi-quantitative rating of the understanding of the information

Audio recordings of the visits were stored on an encrypted hard disk, transcribed, and anonymized by M.C. and a second communication specialist (M.F.L.). Only anonymized transcripts were used for the analyses, which also considered follow-up interview results. M.C. and M.F.L. independently read all transcriptions and systematically labelled different types of misunderstandings and instances of problematic understanding by adopting a coding scheme [18] already tested in previous analyses of patient/doctor interactions [18,19]. The coding scheme offers a method to highlight communication breakdowns, allowing the identification of recurring problems in healthcare communication.

Following the coding scheme [18], seven types of evidence for problematic understanding were considered in the conversational turns. In the context of conversation analysis, a "turn" refers to a unit of speech or communication that constitutes a single participant's contribution to a conversation. It can consist of a single word, phrase, sentence, or longer segment of speech. Turns are the building blocks of conversational interactions, where participants alternate speaking roles to engage in dialogue. The transition from one participant's turn to another often involves cues such as pauses,

tone of voice, or non-verbal signals marking the beginning and end of each turn in a conversation. Understanding the dynamics of turn-taking is essential for the study of discourse and conversation patterns.

Three predefined and mutually exclusive categories of linguistic evidence were considered: strong (lack of understanding, semantic alternative understanding, and pragmatic alternative understanding), acceptable (clarification and checking for understanding), and weak (no uptake or irrelevance). The "strong evidence" category represents actual misunderstandings, while the "acceptable evidence" group indicates doubtful understandings. The "weak evidence" group captures indirect indicators of potential misunderstandings, such as a lack of coherence between the interlocutors' turns. [Supplementary Table 1](#) provides a brief description and example of each coding category. As we were interested in detecting the part of the visit and what kind of problematic understanding occurred, we divided each visit into three main stages: taking patients' personal medical history, diagnosis, and discussion of the next steps (further examinations and treatments). Eventually, we added a further label to each piece of evidence of problematic understanding by assigning one of the three moments of the visit.

Of the 30 patients included, seven (23%) doctor/patient interactions were coded together by M. C. and M. F. L. as training. The remaining 23 doctor/patient interactions were coded independently. There was high agreement between M.C. and M.F.L. regarding evaluation of communication and understanding features (agreement 97.9%; Cohen's Kappa = 0.98) [20]. M.C. and M.F.L. met several times during discussions of doubts and problematic instances. The types of problematic understanding were detected and labelled using MaxQDA22 [21].

2.5. Statistical analysis

Categorical variables were compared using Fisher's or Chi-square tests, and continuous variables were compared using the T-test or Mann-Whitney test, as appropriate. ANOVA and Kruskal–Wallis tests were used for multiple comparisons with Bonferroni correction. Pearson's test was used to evaluate the correlations between continuous variables. An univariable logistic regression analysis with high PHEs® level (3 or 4) and compliance as outcome variables, and patients' disease features and degree of communication as explanatory variables was performed to assess the impact of patient understanding on engagement and compliance. As there have been no previous studies on the association between the patient's understanding rate and the level of engagement and compliance in patients with PDAC, this study was considered a pilot study, and the power size was not calculated.

3. Results

3.1. Patients' features

During the study period, 33 patients were approached to participate. One patient refused to participate, and 32 (96.9%) provided consent. Two of them were subsequently excluded due to screening failure: one as at a re-evaluation of imaging and pathology findings, a diagnosis of cholangiocarcinoma and not PDAC was formulated; the other as the diagnosis was of PDAC recurrence in the remnant pancreas after initial surgery some years before. The demographic and clinical features of the 30 PDAC patients are presented in [Table 1](#). The gender distribution was even (50% female), mean age was 63.7 (95% CI 60.1–67.2). Most patients had locally advanced or metastatic disease, and in only one patient, upfront surgery was the proposed treatment, whereas in the other

Table 1
Patients' demographics and clinical features.

	Total (n = 30)
Age, years - mean (95% CI)	63.7 (60.1–67.2)
Sex, male, n (%)	15 (50%)
Geographic provenance	
Northern Italy	22 (73.3%)
Center-Southern Italy	8 (16.7%)
Educational level	
Illiterate	1 (3.3%)
Primary school	3 (10%)
Middle school	9 (30%)
Secondary school	14 (46.6%)
College degree	3 (10%)
Referral physician	
Oncologist	14 (46.6%)
Gastroenterologist	14 (46.6%)
Surgeon	2 (6.6%)
Resectable	5 (16.6%)
Borderline Resectable	8 (26.6%)
Locally Advanced	4 (13.3%)
Metastatic	13 (43.3%)
First line treatment	
Chemotherapy	29 (96.6%)
Neoadjuvant	12 (40%)
Palliative	17 (56.6%)
Surgical resection	1 (3.4%)
OS, months - mean (95%CI)	12.4 (10.1–14.8)

SD: Standard Deviation.

29 patients, chemotherapy, either with a neoadjuvant or palliative indication, was advised. Overall, 15 (50%) patients died of the disease, with a mean overall survival of 12.4 months (95% CI 10.1–14.8) and a median follow-up time of 19 months (range 7–24).

3.2. Patients' engagement and compliance

The PHE-s® results address an overall satisfying engagement as a high PHE level (3 or 4) was recorded in 21/30 (70%) patients and only 3 (10%) were in “Blackout.” However, the PHE-s® results did not correlate with the patient variables. In fact, the mean level of engagement was 2.93/4, being non-significantly higher in patients with a high education level (high school or college) vs. low (3.1 vs 2.8; $p = 0.45$), in female patients vs. male (3 vs. 2.8; $p = 0.79$), in patients seen by a gastroenterologist vs. oncologist (3.5 vs 2.9; $p = 0.44$), non-metastatic vs. metastatic disease (3.1 vs 2.8; $p = 0.49$), and from the Center/South vs. North of Italy (3.4 vs. 2.7; $p = 0.19$).

Nonetheless, while 100% (8/8) of patients from the Center and South of the country had a high PHEs® level, this was the case for only 59% (13/22) of the patients from the North ($p = 0.06$). The mean age of patients with high (62.5; 95% CI 58.1–66.9) or low (66.3; 95% CI 59.3–73.3) PHEs® levels did not differ significantly ($p = 0.32$), and age did not correlate with PHEs® levels ($\rho = -0.043$; $p = 0.82$). All but one patient who refused the proposed treatment and asked for a second opinion at another center were compliant (96.6%).

3.3. Evaluation of the visits in terms of communication and understanding

The mean time of the recordings of the visits was 31.1 min (95% CI 26.9–35.4) with a significantly longer duration in the oncology visits vs. gastroenterology or surgery visits (mean 37.8 vs 26.2 vs 16 min; $p = 0.02$).

There were 6780 conversational turns, 3431 (50.6%) doctors,

2537 (37.4%) patients, and 812 (12%) caregivers. The median number of conversational turns was 121 for doctors, 75 for patients, and 20 for caregivers ($p < 0.0001$, Kruskal–Wallis test for multiple comparisons, no difference between doctors and patients) (Fig. 1, panel A). The difference in terms of time of participation in the conversation was, however, more relevant when examining the percentage of time speaking during the visit, with a median of 77% of the total time for doctors, 13% for patients, and 2% for caregivers ($p < 0.0001$ Kruskal–Wallis for multiple comparisons, significant difference between all groups). (Fig. 1, Panel B).

Next, we analysed the number of turns and their lengths based on the patient's sex. While doctor-patient conversations had a mean of 233 (183–283) conversational turns, the mean number of turns was 94.2 for female patients (41%) vs 80.2 for male patients (30%) ($p = 0.5$). Considering the time of visit, the median percentage of conversations held by the patients was 14% (7–20) for female patients and 9.5% (5–16) for male patients ($p = 0.43$).

There was no significant difference in terms of the median percentage of conversational time between patients from the North or Center/South ($p = 0.28$), while patients with a high educational level (secondary school or college 14% vs. 6% lower levels; $p = 0.12$) and metastatic (16% vs. 8.5% non-metastatic) displayed a significantly higher median percentage of conversational time. In addition, there was no correlation between the median conversational time and patient age ($\rho = -0.16$; $p = 0.30$).

Because we were interested in the role of caregivers, we investigated whether their gender had an influence. For all but four patients, the sex of the caregiver was opposite to that of the patients. Caregivers tended to be family members, being either the spouse/partner (40%) or the son/daughter (23.3%) in most cases; three patients were accompanied by their mothers, sisters, or friends. In the interactions where the patients had a female caregiver, the patients had a mean of 95 conversational turns (median 11.6% of the total length of the visit) and their caregivers had a mean of 56 (median 6% of the visit). Conversely, patients with male caregivers had a mean of 73 turns (13% of the time), and their caregivers had a mean of 11.7 (only 1.3% of the visit) (chi-square $p = 0.06$) (Fig. 2A and B). These data suggest that female caregivers intervene more often and for a longer time during doctor-patient interviews than their male counterparts do.

3.4. Evaluation of understanding

In the 30 doctor-patient interviews that were analysed, there were 290 instances of problematic evidence, which accounted for 4.28% of the total 6780 conversational turns. This rate varied widely, the median being 2.9% (IQR 0.81%–6.66%) with a wide range of 0.4%–22%, and only two visits in which the rate of conversational turns with instances of misunderstandings was $\geq 10\%$ of the conversational turns.

Table 2 highlights the different instances of identified problematic evidence and their distributions among doctors, patients, and caregivers. Following the theoretical distinction by Rossi & Macagno [13], we analysed separately the cases of “strong evidence”, “acceptable evidence” and “weak evidence.” As an example, the cases of “acceptable evidence” were much more frequent among doctors (chi square $p < 0.0001$) while “strong evidence” (most relevant misunderstandings) cases were not.

As for the time of visit when problematic understanding occurred, there were also significant differences between doctors and patients/caregivers. Indeed, while we found that for doctors most of these instances happened during the taking of patients' personal medical history, for patients and caregivers, most

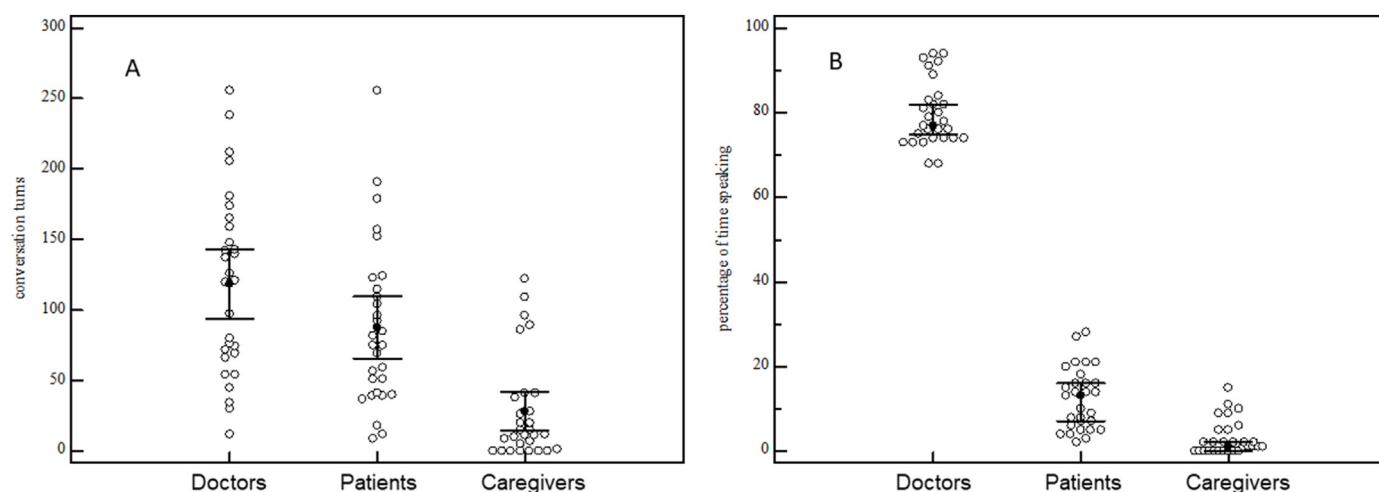


Fig. 1. **Panel A:** Number of conversational turns. The median number of conversational turns was 121 for doctors, 75 for patients, and 20 for caregivers ($p < 0.0001$ Kruskal–Wallis for multiple comparisons, but no difference between doctors and patients); **Panel B:** percentage of time speaking. The median percentage was 77% of the total time for doctors, 13% for patients, and 2% for caregivers ($p < 0.0001$ Kruskal–Wallis for multiple comparisons, with a significantly higher percentage for doctors than for both patients and caregivers).

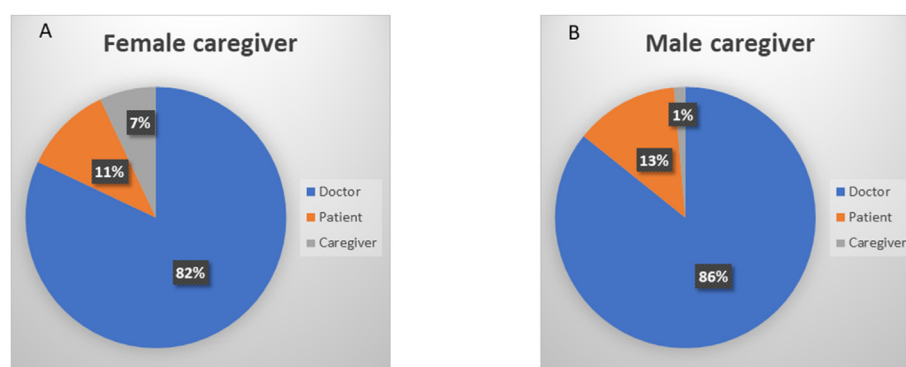


Fig. 2. Percentage of time speaking according to sex of caregivers. In the interactions where the patients had a female caregiver, the patients had a median conversational time of 11.6% of the total length of the visit and their caregivers of 6%. Conversely, patients with male caregivers had a median of 13% of the time, and their caregivers had only 1.3% of the visit (Chi square, $p = 0.06$).

problematic understandings occurred during the discussion of diagnosis or “next steps” (Table 3; chi square $p < 0.0001$).

3.5. Follow-up interviews

While the linguistic analysis displayed cases of problematic understanding, follow-up interviews did not highlight patients' perceptions of problems in communicating with their physicians. During the post-visit interviews that took place over the phone, almost no patients reported problems understanding during the diagnostic visit. Instead, most of them highlighted their physicians' competences and professionalism. Two patients explicitly reported issues in the interaction with the physician; more specifically, one patient complained about a lack of empathy in using technical language, and another about the difficulty of asking questions to the physician.

3.6. Association between patients' and communication features and compliance PHEs®

As all patients, except one, were compliant with the proposed treatment, analyses regarding compliance were not feasible. An univariable logistic regression analysis was performed to

investigate the association between patient and conversation features and level of engagement according to the PHEs® scale (Table 4). Interestingly, in this analysis the only factor associated with high PHEs® levels was an origin from central and southern Italy instead of northern Italy ($p = 0.0087$).

4. Discussion

This is the first study to evaluate the quality of communication and level of engagement in the setting of initial care for PDAC patients. A semi-quantitative approach was employed, with engagement measured through a validated scale [12], and the features of the communication were analysed through a tool whose inter-coder reliability was already strong in previous studies [13].

The conversation time varied widely and was significantly longer when the physician in charge was an oncologist. This may be due to the more detailed explanation of chemotherapy and its side effects. In addition, as expected, doctors spoke most of the time and posed more questions than did patients and caregivers. There were differences in doctor–patient interactions based on the gender of the patients and caregivers, as female patients and caregivers tended to have more and longer turns than male patients and caregivers (see Fig. 2A and B). However, these communication

Table 2
Heatmap highlighting the distribution of observed misunderstandings among Doctors, Patients and Caregivers and their type. The red colour indicates the group with the highest rate of misunderstanding for each category, the green the group with the lowest rate of misunderstanding and the yellow the intermediate group.

	Strong Evidence (Lack of understanding + Semantic alternative understanding + Pragmatic alternative understanding)		Acceptable Evidence (Clarification + Check)		Weak Evidence (No uptake + Irrelevance)		total	
	N	%	%	%	n	%	n	
Doctors	16	5.5	161	55.5	12	4.1	189	p<0.0001
Patients	20	6.8	52	17.9	5	1.7	77	p<0.0001
Caregivers	6	2.0	18	6.2	0	0.0	24	p=0.024
total	42		231		17		290	
		p=0.38			p<0.0001			p=0.0017
					p=0.33			

Table 3
Moments of the visits when the misunderstandings occurred.

	Doctors	Patients & Caregivers
Taking of patients' personal medical history	152	3
Diagnosis	2	22
Next Steps (Further tests and treatment)	2	55

Chi square $p < 0.0001$.

variables were not associated with patient engagement or compliance.

The levels of compliance and engagement were generally high, but in some patients, the PHEs scale highlighted a blackout condition. Among the many clinical and demographic factors, only the region of origin of the patient seemed to be associated with engagement, whereas conversion-related variables were not. There are several possible explanations for this observation: First, while traditionally in the South of Italy, there is a more conservative culture and social representation of the healthcare system, this is often coupled with more barriers in access to high-quality healthcare, including for PDAC [22]. Thus, access to care requires more motivation and effort for patients from Center/South Italy who travelled longer to reach a tertiary expert center located in the North. This likely affected their attitudes towards the care pathway and doctor-patient interactions, resulting in higher engagement. The absence of other variables associated with the level of engagement may be a type II error, owing to the relatively small sample size. As for the conversation, there was a discrepancy between the number of turns and their length: physicians tended to have longer turns than did patients and caregivers (Fig. 1). When individuals engage in lengthy turns during a conversation or any form of communication, it often suggests that the information being conveyed is more intricate or detailed. Although this complexity can be valuable, it can also pose difficulties to listeners.

Longer turns may demand a greater cognitive load to process and comprehend, potentially leading to comprehension problems.

It is interesting to note that, in the present study, female companions/caregivers often took on the role of keeping medical records and actively participating in their partner/parent's medical treatment. This finding is consistent with previous research that suggests that women often take on a caregiving role in families when a person is diagnosed with severe illness [23,24]. Overall, the findings of this qualitative analysis support the idea that female companions/caregivers play an important role in supporting care recipients during illness.

As the actual understanding of received information can be a problematic issue for patients, especially during a stressful event, such as the first communication of a potentially lethal disease, we employed a previously validated coding tool to evaluate the problems of communication in a semi-quantitative manner. In this view, it is worth noting that the labels Clarification and Check of Understanding should be viewed as moments indicating potential problems solved by deepening the interaction between the participants, thus facilitating mutual understanding. Clarification and Check of understanding are normal moments of the complex communicative interaction between cancer patients and their physicians; for instance, the Roter Interaction Analysis System classifies them as check mechanisms aimed at promoting accuracy, recall, and understanding [25]. By comparing our results with those of similar studies that employed the same coding tool (Supplementary Table 2), we found that our data are consistent with those of previous research.

An important part of the study was to conduct a follow-up evaluation between a communication specialist and patients to check the clarity of the information received by their physicians. None of the patients reported any problematic understanding during the diagnostic visit despite the problematic evidence identified in this study. This phenomenon is also confirmed by the

Table 4
Logistic Regression of Factors associated with a High PHE level.

	OR (95% CI)	p value
Age	0.95 (0.87–1.04)	0.30
Female Gender	1.37 (0.28–6.6)	0.69
Educational Level		
Illiterate or Primary School	1 (reference)	–
Middle School	0.3 (0.07–2)	0.2
Secondary School	4.6 (0.77–28)	0.07
College	0.17 (0.01–2.2)	0.1
Stage of disease		
Resectable/Borderline resectable	1 (reference)	–
Locally Advanced	0.36 (0.04–3.1)	0.33
Metastatic	1.81 (0.35–9.27)	0.46
Geographic Area of Italy		
North	1 (reference)	–
Center or South	98.9 E+006	0.0087
Specialty of Physician		
Oncologist	1 (reference)	–
Gastroenterologist or Surgeon	1.66 (0.34–8.3)	0.52
Length of the visit (per increasing minutes)	1.01 (0.93–1)	0.76
Rate of Conversational turns of the patient		
<5%	1 (reference)	–
>20%	0.5 (0.06–3.74)	0.50
Number of conversational turns of the patient (per increasing number)	1 (0.99–1.02)	0.29
Misunderstandings of the patients		
None	1 (reference)	–
At least one	0.72 (0.15–3.49)	0.69
Questions from the patient		
None	1 (reference)	–
At least one	0.39 (0.07–2)	0.26

literature and falls under the trend discourse [26] and is labelled as masking because patients tend not to report any problems and behave in the way they feel supposed to behave. On the other hand, it is possible that despite imperfect understanding, patients still trusted their treating doctor; thus, they did not report problems during the follow-up interview.

The present study has the following strengths: (a) it is the first study on this topic; (b) the study design was rigorous and included several established phases and the use of validated tools; and (c) the research team that designed and conducted the study had multidisciplinary competence, including different specialists with specific expertise on PDAC, communication specialists, and psychologists. However, this study has some limitations, such as the relatively small sample size, the lack of an *a priori* power calculation, and the possible occurrence of other non-investigated factors that may account for the observed heterogeneity regarding conversation, understanding, and engagement. Second, because all but one patient accepted the proposed treatment plan and was considered compliant, it was not possible to verify whether engagement and communication influenced compliance. Indeed, in deadly diseases such as PDAC, where the most prescribed treatment is intravenous chemotherapy, it is uncommon for patients to refuse it, especially if it is proposed in a tertiary center, and it is difficult to directly measure compliance. Alternative tools to measure compliance, such as interviews and administration of self-report likert scales [27] may prove useful in such settings. Applications (apps) may also be particularly useful, as reported by preliminary findings in patients undergoing pancreatic surgery [28]. Finally, all the physicians leading the visits were male, at times supported by younger female colleagues. Existing literature suggests that there may be differences in communication styles and the way these are perceived between male and female physicians [29,30]. In addition, all the involved doctors were experts and committed to treating PDAC. Thus, future studies on this topic will require larger and better-balanced groups of male and female

physicians with different degrees of experience working in different settings. This would lead to the identification of recurring topics that are problematic in doctor-patient interactions and allow the development of guidelines for quality communication. For a future study, it would be interesting to include a follow-up interview with patients in the middle of the treatment to check if their perspectives are coherent with the first ones collected, if they have the chance to change their minds, and if their engagement is still high or low.

The present study represents the first attempt to measure clarity of communication and engagement in patients with PDAC. None of the investigated variables, but the origin of the patient, explained the variability in the engagement level. However, we were able to report in detail the characteristics of misunderstandings during communication between treating physicians and patients with PDAC. As communication failures are frequently reported by cancer patients [31], it would be interesting to further investigate the trajectory of engagement of PDAC patients along their entire disease journey and how this is influenced by the clarity of communication.

Authors' contribution

Conceptualization: MC, GC, GG, CM; Data curation: MC, MFL, GC, NP, MM, MR, LA, MF; Formal analysis: MC, MFL, GC, MGR; Funding acquisition: GC; Investigation: MC, MFL, GC, MM, MR, LA, MGR; Methodology: MC, MFL, CM, GC, MF, MR, PGA, MGR, GG; Project administration: MC, GC; Resources: GC; Software: GC; Supervision: GC, CM, MGR, GG; Roles/Writing -original draft: MC, MFL, CM, GC, LA; and Writing - review & editing: all authors. All authors made substantial contributions to the conception and design of the study, acquisition, analysis and interpretation of data, drafting the article or revising it critically for important intellectual content and all approved the version to be submitted.

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Appendix A. Supplementary data

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